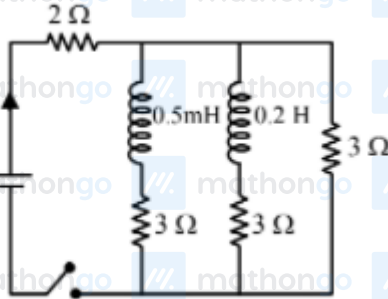


Q1 2021 (01 Sep Shift 2)

For the given circuit the current i through the battery when the key is closed and the steady state has been reached is



(1) 6 A

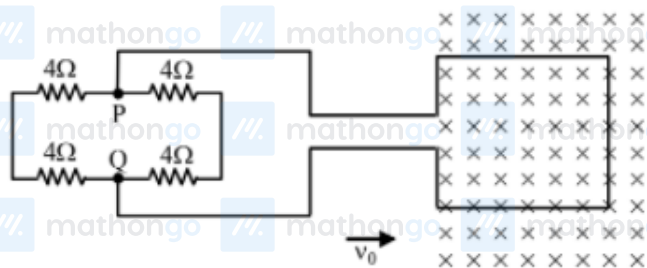
(2) 25 A

(3) 10 A

(4) 0 A

Q2 2021 (01 Sep Shift 2)

A square loop of side 20 cm and resistance 1Ω is moved towards right with a constant speed v_0 . The right arm of the loop is in a uniform magnetic field of 5 T. The field is perpendicular to the plane of the loop and is going into it. The loop is connected to a network of resistors each of value 4Ω . What should be the value of v_0 so that a steady current of 2 mA flows in the loop?



(1) 1 m/s

(2) 1 cm/s

(3) 10^2 m/s(4) 10^{-2} cm/s

Q3 2021 (31 Aug Shift 2)

A coil is placed in a magnetic field $\vec{B}$ as shown below :

- (1) Outward and decreasing with time
- (2) Parallel to the plane of coil and decreasing with time
- (3) Outward and increasing with time
- (4) Parallel to the plane of coil and increasing with time

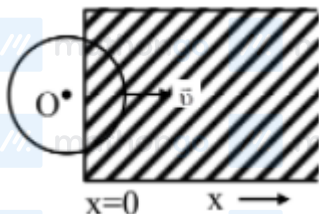
Q4 2021 (31 Aug Shift 1)

A small square loop of side 'a' and one turn is placed inside a larger square loop of side b and one turn ($b \gg a$). The two loops are coplanar with their centres coinciding. If a current I is passed in the square loop of side 'b', then the coefficient of mutual inductance between the two loops is :

- (1) $\frac{\mu_0}{4\pi} 8\sqrt{2} \frac{a^2}{b}$
- (2) $\frac{\mu_0}{4\pi} \frac{8\sqrt{2}}{a}$
- (3) $\frac{\mu_0}{4\pi} 8\sqrt{2} \frac{b^2}{a}$
- (4) $\frac{\mu_0}{4\pi} \frac{8\sqrt{2}}{b}$

Q5 2021 (27 Aug Shift 2)

A constant magnetic field of 1 T is applied in the $x > 0$ region. A metallic circular ring of radius 1m is moving with a constant velocity of 1 m/s along the x -axis. At $t = 0$ s, the centre of O of the ring is at $x = -1$ m. What will be the value of the induced emf in the ring at $t = 1$ s ? (Assume the velocity of the ring does not change.)



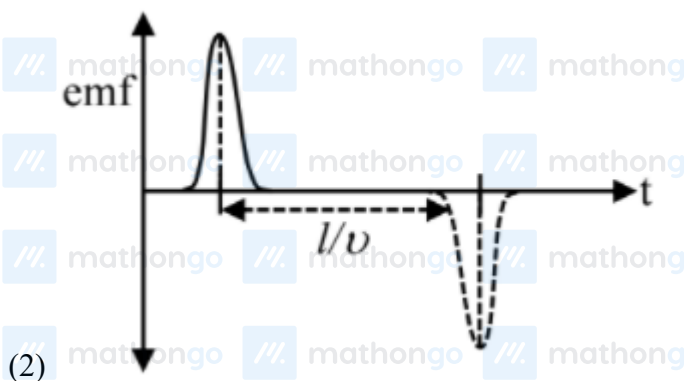
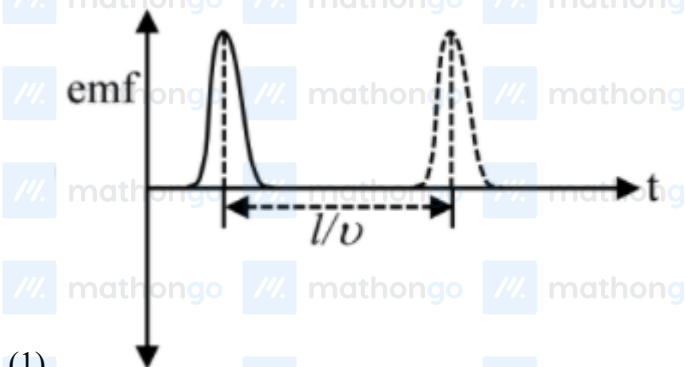
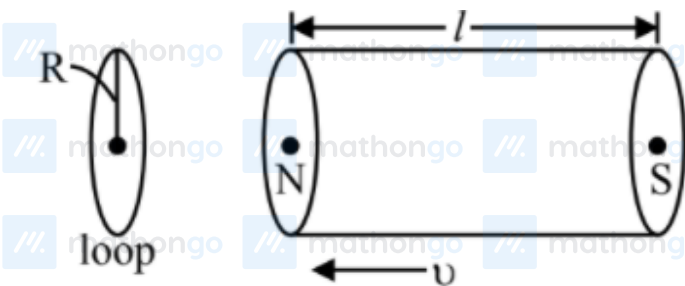
- (1) 1 V
- (2) 2π V

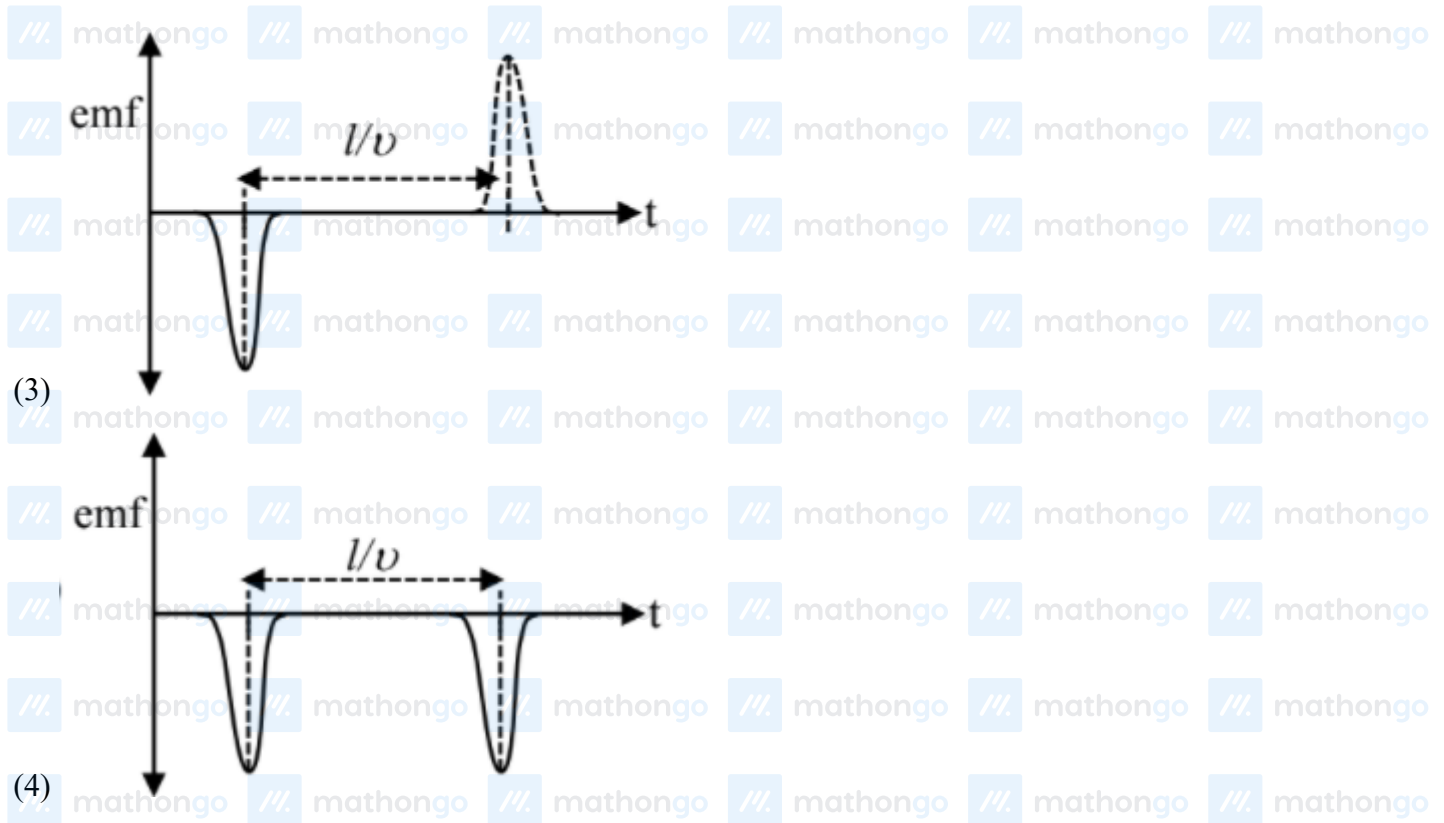
(3) 2 V

(4) 0 V

Q6 2021 (27 Aug Shift 1)

A bar magnet is passing through a conducting loop of radius R with velocity v . The radius of the bar magnet is such that it just passes through the loop. The induced e.m.f. in the loop can be represented by the approximate curve :





Q7 2021 (26 Aug Shift 2)

A circular coil of radius 8.0 cm and 20 turns is rotated about its vertical diameter with an angular speed of 50rad s^{-1} in a uniform horizontal magnetic field of 3.0×10^{-2} T. The maximum emf induced the coil will be $\dots \times 10^{-2}$ volt (rounded off to the nearest integer)

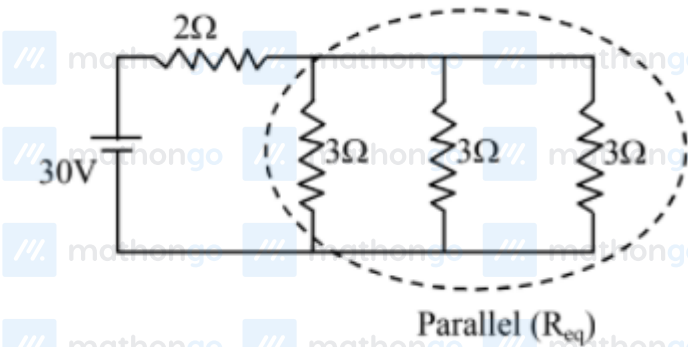
Answer Key

Q1 (3)	Q2 (2)	Q3 (1)	Q4 (1)
Q5 (3)	Q6 (3)	Q7 (60)	

Q1 (3)

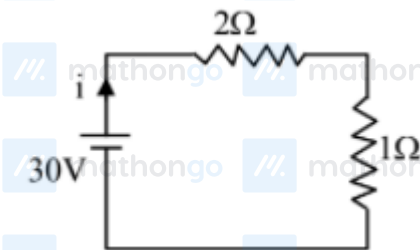
In steady state, inductor behaves as a conducting wire.

So, equivalent circuit becomes



$$\frac{1}{R_{eq}} = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$$
$$\Rightarrow R_{eq} = 1\Omega$$

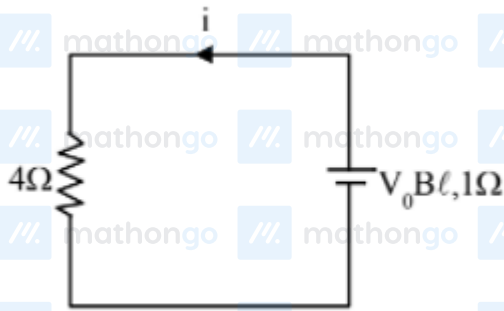
$\Rightarrow$ Circuit becomes



$$\Rightarrow i = \frac{30}{3} = 10 \text{ A}$$

Q2 (2)

Equivalent circuit



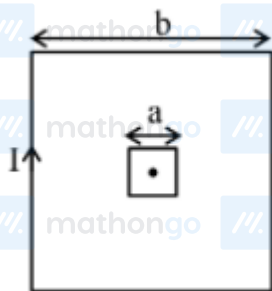
$$i = \frac{V_0 B \ell}{4 + 1} \Rightarrow V_0 = \frac{5(2 \text{ mA})}{5 \times .2} = 10^{-2} \text{ m/s} = 1 \text{ cm/s}$$

Option (2)

Q3 (1)

$\vec{B}$ must not be parallel to the plane of coil for non zero flux and according to lenz law if B is outward it should be decreasing for anticlockwise induced current.

Q4 (1)

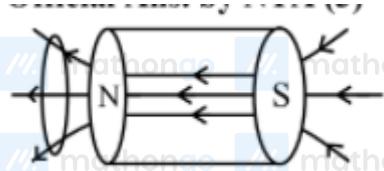


$$B = \left[\frac{\mu_0 I}{4\pi} \frac{1}{b/2} \times 2 \sin 45 \right] \times 4 \phi = 2\sqrt{2} \frac{\mu_0 I}{\pi} \frac{1}{b} \times a^2 \therefore M = \frac{\phi}{I} = \frac{2\sqrt{2}\mu_0 a^2}{\pi b} = \frac{\mu_0}{4\pi} 8\sqrt{2} \frac{a^2}{b} \text{ Option (1)}$$

Q5 (3)

$$\begin{aligned} \text{emf} &= BLV \\ &= 1. (2R) \cdot 1 \\ &= 2 \text{ V} \end{aligned}$$

Q6 (3)



→ When magnet passes through centre region of solenoid, no current / Emf is induced in loop.

→ While entering flux increases so negative induced emf

→ While leaving flux decreases so positive induced emf.

Q7 (60)

Maximum emf $\varepsilon = N\omega AB$

$$N = 20, \omega = 50, B = 3 \times 10^{-2} \text{ T}$$

$$\varepsilon = 20 \times 50 \times \pi \times (0.08)^2 \times 3 \times 10^{-2} = 60.28 \times 10^{-2}$$

Rounded off to nearest integer = 60

Ans. 60